

Q1: Multiple choice questions

(2pts)

1. "Software as a Service (SaaS)" slides list one benefit as pay-as-you-go and accessibility, and one drawback as:

- ☐ Guaranteed performance ☐ Full developer control ☒ Security/trust vulnerabilities and internet dependence ☐ Free on-prem hardware

2. According to "Application types," what determines which engineering methods/tools to use?

- ☐ Project manager's personal preference ☒ Application type, customer requirements, team background, and tools ☐ Software cost alone ☐ All of the above

3. A key weakness of the Waterfall model in the slides is:

- ☐ Low documentation ☐ Iterative ☐ Lack of any structure ☒ Inflexible to changing requirements and late problem discovery

4. To cope with change, there are two approaches

- ☐ Code-and-fix & Big-bang integration ☐ Change prevention & big-bang integration ☒ Prototyping & incremental delivery ☐ Waterfall with extended design phase & late verification

Q2: For each statement below, classify (i) User vs. System requirement and (ii) Functional (FR) vs. Non-functional (NFR). (4pts)

Requirement	User vs. System	FR vs. NFR
"The system should be easy to use for first-year students."	[redacted]	[redacted]
"The application shall allow password reset using a verified email."	[redacted]	[redacted]
"The report generation shall be fast."	[redacted]	[redacted]
"The interface will display student GPA to two decimal places."	[redacted]	[redacted]

Q3: Team of 6 people; workday = 8h. Last 4 sprints completed story points: 22, 28, 26, 24.

1. Compute the sprint velocity (use the average sprint story points).

[redacted]

[redacted]

2. Compute focus factor ($\text{SprintVelocity} \div (\text{people} \times \text{hours/day})$).

(1pts)

[redacted]

[redacted]

3. For a 3-week sprint (Mon-Fri, weekends off), compute effective team capacity (hours) using the focus factor. Show formulas and round sensibly.

(1pts)

[redacted]

[redacted]

[redacted]

[redacted]

4. If the next sprint backlog is 30 story points, state whether the plan is feasible and why.

Feasibility: ☒ Feasible

or

☐ Prudent

(1pts)

Justification:

[redacted]

[redacted]

[redacted]

[redacted]

Q4: For each project, choose one development process model: Agile, Plan-Driven, or Component-Based. (4pts)

Project 1 — Smart-City Sensor Hub

- **Stakeholder Presence:** Moderate; city IT holds bi-weekly reviews; end-users (operators) rarely available.
- **Requirement Volatility:** Medium; new sensor types/vendors appear mid-project, but core data collection goals are stable.
- **Project Size & Team:** 20 engineers across 3 vendors; distributed.
- **Architecture Constraints:** Must integrate COTS gateways and vendor SDKs; strict uptime (99.9%).
- **Compliance:** Must pass basic cybersecurity hardening but no heavy certification.

Choose: ☐ Agile ☒ Plan-Driven ☒ Component-Based

Project 2 — Banking Regulatory Reporting

- **Stakeholder Presence:** Limited; auditors and compliance office set requirements up front and review only at stage gates.
- **Requirement Volatility:** Low; templates and KPIs fixed by regulation for the next 3 years.
- **Project Size & Team:** 30 people, multi-department; formal PMO.
- **Documentation & Traceability:** Detailed SRS/SDD, audit trails, sign-offs required.
- **Risk Profile:** High penalty for defects; schedule predictable.

Choose: ☐ Agile ☒ Plan-Driven ☒ Component-Based

Project 3 — Campus Social-Learning MVP

- **Stakeholder Presence:** High; product owner and pilot students give daily feedback.
- **Requirement Volatility:** High; features frequently reprioritized to find product-market fit.
- **Project Size & Team:** 6 developers, co-located; CI/CD in place.
- **Time to Market:** Need public beta in 4-6 weeks; success measured by adoption/engagement.
- **Architecture:** Greenfield; minimal legacy constraints.

Choose: ☒ Agile ☐ Plan-Driven ☐ Component-Based

Project 4 — Medical Device Companion App

- Stakeholder Presence: Moderate; clinicians available for periodic usability reviews; regulatory team sets constraints.
- Requirement Volatility: Medium; UI iterations expected, but clinical workflows and data formats are fixed by standards.
- Project Size & Team: 15 people; UX + mobile + back-end.
- Compliance: Must align with IEC 62304 software life-cycle documentation and ISO 14971 risk management; app is Class II accessory.
- Integration: Uses a certified SDK from the device vendor for data acquisition.

Choose: ☐ Agile ☒ Plan-Driven ☐ Component-Based

Q5: For each statement, indicate the requirement type: Product, Organization or External (6pts)

1. The API shall respond in ≤ 300 ms (p95) for read requests under 100 concurrent users.

Choose: ☒ Product ☐ Organization ☐ External

2. All new services must be built in Java 21 and deployed on the organization's Kubernetes platform.

Choose: ☐ Product ☒ Organization ☐ External

3. Patient data must comply with HIPAA and local privacy law X.

Choose: ☐ Product ☐ Organization ☒ External

4. The mobile app shall support offline read-only access for 24 hours.

Choose: ☒ Product ☐ Organization ☐ External

5. The system shall export reports in CSV and PDF.

Choose: ☐ Product ☐ Organization ☐ External

6. Device firmware must interoperate with ISO/IEC 14443 Type A NFC readers.

Choose: ☐ Product ☐ Organization ☐ External